Road Inspector: An AI-Powered Road Quality Inspection and Repair Estimation System Using Computer Vision, Semantic Segmentation and Predictive Analytics

***Abstract*—Road infrastructure maintenance remains challenging because traditional manual inspection can cause delayed defect identification, inconsistent assessment, and inefficient maintenance planning. This research proposes Road Inspector, an AI-powered road quality inspection and repair estimation ecosystem integrating computer vision, deep learning, semantic segmentation, predictive analytics, and environmental intelligence. The system uses YOLOv8 and Convolutional Neural Networks for road defect detection, U-Net and semantic segmentation for repair area estimation, Random Forest models for accident risk and material estimation, and environmental analysis for repair scheduling. Beyond defect detection, it provides severity assessment, repair area calculation, accident risk prediction, material estimation, tar heating optimization, and environmental suitability analysis. Experimental results indicate that the proposed system can support data-driven road maintenance decisions, resource allocation, repair planning, and road safety.**

***Index Terms*—Artificial Intelligence, Computer Vision, Predictive Analytics, Road Quality Assessment, Semantic Segmentation, YOLOv8.**

I. Introduction

Road transportation infrastructure supports economic activity, public transportation, logistics, and social development. Road surfaces are continuously affected by environmental conditions, traffic loads, and natural wear, producing potholes, longitudinal cracks, transverse cracks, alligator cracks, edge failures, and surface erosion. If these defects are not detected and repaired early, they can increase maintenance costs, vehicle damage, traffic disruption, accident risk, and public-safety concerns.

Traditional inspections rely mainly on field engineers and maintenance personnel. These processes are labor-intensive, time-consuming, costly, and dependent on human judgement, which can lead to delayed reporting, inconsistent assessments, limited scalability, and inefficient resource allocation. As road networks expand, automated and intelligent monitoring becomes increasingly important.

Recent advances in Artificial Intelligence (AI), Computer Vision, Deep Learning, and Internet of Things (IoT) technologies provide new opportunities for automated road assessment. Existing studies have applied machine learning and computer vision to road defect detection [1]–[4], [8], [9]. CNNs and object detectors can identify road damage from images and video, but many systems remain focused on detection and classification rather than maintenance decision support [5], [6]. Severity estimation, repair area calculation, accident risk, material requirements, environmental suitability, and repair scheduling are still insufficiently integrated.

Road Inspector addresses this gap by combining YOLOv8 and CNN-based defect detection, U-Net semantic segmentation, Random Forest predictive analytics, and environmental analysis in one platform. Its objective is to detect road defects, estimate severity, predict accident risk, calculate repair area and materials, and support repair scheduling. The main contributions are:

1) AI and IoT-based road defect detection using YOLOv8 and computer vision.
2) Semantic segmentation-based repair area estimation for precise damage boundaries.
3) Intelligent material and heat estimation for repair planning and resource optimization.
4) Environmental and traffic analysis for maintenance scheduling and decision support.
5) Integration of the modules into a unified inspection and reporting ecosystem.

The remainder of this paper reviews related work, presents the system architecture and intelligent modules, reports experimental results, and concludes with limitations and future directions.

II. Literature Review

This section reviews research related to road defect detection, semantic segmentation, material estimation, and maintenance planning.

A. AI and IoT-Based Road Defect Detection

Rathee et al. [1] reviewed AI methods for detecting potholes, cracks, and road anomalies, showing the potential of automated inspection for road monitoring. Alrajhi et al. [2] examined IoT sensors, vibration analysis, accelerometers, and machine learning for real-time road condition assessment. Image-based methods have also become common [3], [4], [7]–[9], [13]. Swetha and Devnath [3] used Mask R-CNN for pothole and

crack localization, Maeda et al. [8] developed a smartphone-based CNN framework for multiple road-damage classes, and Lei et al. [9] applied deep CNNs to crack detection. Together, these studies support the use of computer vision and deep learning for automated defect identification.

B. Repair Area Estimation Using Semantic Segmentation

Bounding boxes do not represent irregular damage boundaries accurately, motivating pixel-level segmentation approaches [10]–[12], [14]–[17]. Katsamenis et al. [10] proposed a ResUNet-a framework for road-damage segmentation, while Fan et al. [11] improved pothole boundary extraction through multi-scale feature fusion. Tseng et al. [12] combined RGB and LiDAR data; Kumar et al. [16] demonstrated U-Net for potholes and cracks; and Fei et al. [17] used an encoder-decoder network for crack boundaries. These approaches show that semantic segmentation is suitable for repair boundary and area estimation.

C. Automated Material Requirement Estimation

Material estimation can reduce wastage and improve maintenance planning. Zaimis et al. [18] combined drone-based pothole detection with volume calculation and reported approximately 90% accuracy, while Bruno et al. [19] converted photogrammetrically measured pothole volumes into repair material quantities. Other studies used stereo vision, point clouds, Structure from Motion, LiDAR, depth maps, and 3D reconstruction for pothole dimensions and volume estimation [20]–[26]. These methods demonstrate the value of geometry and depth information for material allocation.

D. Environmental and Traffic Analysis

Weather, humidity, rainfall probability, temperature, and traffic density influence repair quality and scheduling. Predictive analytics and environmental monitoring can therefore support maintenance periods that reduce disruption. However, these capabilities are frequently implemented independently rather than as part of a unified road-maintenance platform.

E. Research Gap

Existing work provides effective components for defect detection, segmentation, and volume estimation, but often addresses them separately [1], [10], [18]. Integrated support for severity, accident risk, repair area, material and heat estimation, environmental suitability, and scheduling remains limited. Some approaches also depend on LiDAR, stereo cameras, or drones, which can constrain practical deployment [12], [20], [21], [26]. Road Inspector therefore integrates these functions into one intelligent maintenance ecosystem.

III. Proposed System Architecture

The proposed system, Road Inspector, is designed as an intelligent AI-powered road quality inspection and repair estimation ecosystem that automates the complete road maintenance workflow. The architecture integrates Computer Vision,

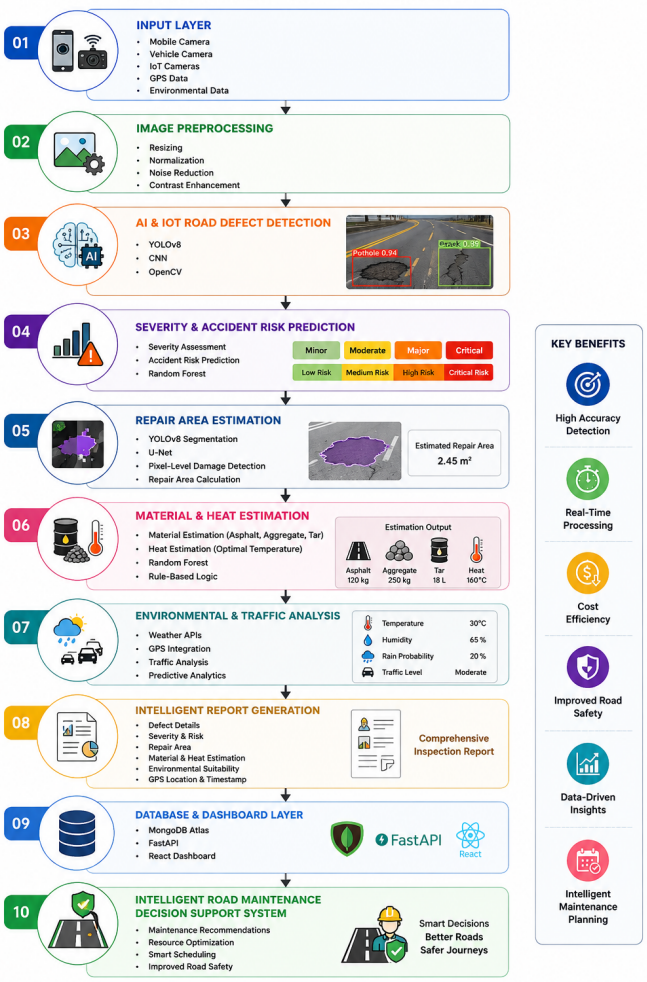


Fig. 1. Proposed Road Inspector system architecture.

Deep Learning, Semantic Segmentation, Predictive Analytics, Environmental Intelligence, and IoT-based data collection into a unified decision-support platform. Unlike conventional road inspection systems that focus solely on defect detection, the proposed framework provides comprehensive maintenance intelligence including severity assessment, accident risk prediction, repair area estimation, material requirement calculation, heat estimation, and repair scheduling optimization.

The overall system architecture consists of four major intelligent modules: AI and IoT-Based Road Defect Detection, Repair Area Estimation, Material and Heat Estimation, and Environmental and Traffic Analysis. These modules operate collaboratively to transform raw road inspection data into actionable maintenance recommendations.

The system begins with the acquisition of road images through mobile devices, vehicle-mounted cameras, or IoT-enabled camera systems. Along with image data, additional information such as GPS coordinates, timestamps, and environmental conditions are collected to support location-based analysis and reporting. The captured images undergo preprocessing operations including image resizing, normalization, noise reduction, and contrast enhancement to improve image quality and detection performance.

The processed images are then forwarded to the AI and IoT-Based Road Defect Detection Module. This module utilizes YOLOv8 and Convolutional Neural Networks to identify road defects such as potholes, longitudinal cracks, transverse cracks, alligator cracks, edge failures, and surface erosion. The detection process generates defect classifications, bounding boxes, and confidence scores while simultaneously estimating the severity level of each detected defect. A Random Forest model is further employed to predict accident risk levels based on defect characteristics and environmental factors.

Following defect detection, the identified damaged regions are passed to the Repair Area Estimation Module. This module employs YOLOv8 Segmentation and U-Net semantic segmentation techniques to perform pixel-level classification of damaged areas. Unlike traditional object detection methods, semantic segmentation accurately identifies irregular defect boundaries and generates detailed damage masks. Through pixel-to-real-world calibration, segmented regions are converted into approximate physical dimensions, enabling accurate repair area estimation and maintenance planning.

The output from the segmentation module is then processed by the Material and Heat Estimation Module. Using Random Forest predictive models and rule-based estimation algorithms, the system calculates asphalt quantity, aggregate requirements, tar requirements, repair material volume, and optimized heating temperature ranges. These estimations are derived from defect dimensions, severity levels, environmental conditions, and road surface characteristics. This module aims to reduce material wastage, improve budgeting accuracy, and optimize repair resource allocation.

The Environmental and Traffic Analysis Module further enhances maintenance decision-making by evaluating weather conditions, humidity levels, temperature data, rainfall probability, and traffic patterns. Based on these factors, the system identifies suitable repair periods, avoids peak traffic hours, and supports intelligent maintenance scheduling. Future enhancements may incorporate route optimization and alternative traffic recommendations to minimize disruption during repair operations.

Finally, all processed information is integrated into an Intelligent Report Generation Module. The system automatically generates comprehensive maintenance reports containing defect type, confidence score, severity level, accident risk assessment, estimated repair area, material requirements, repair recommendations, environmental suitability analysis, GPS coordinates, and timestamp information. These reports provide road maintenance authorities with actionable insights that support efficient planning and resource management.

IV. AI and IoT-Based Road Defect Detection

The AI and IoT-based road defect detection module is the first major intelligent component of the proposed Road Inspector system. The primary objective of this module is to automatically identify and classify visible road surface defects using image-based data collected from mobile devices, vehicle-mounted cameras, and IoT-enabled camera systems. The module focuses on detecting common pavement damages such as potholes, longitudinal cracks, transverse cracks, alligator cracks, edge failures, and surface erosion.

The detection process begins with road image acquisition. Images are captured from road surfaces and combined with supporting metadata such as GPS coordinates, timestamps, and environmental conditions. These data elements allow the system to associate detected defects with their exact location and inspection time, enabling more effective maintenance tracking and report generation.

Before defect detection is performed, the captured images are passed through an image preprocessing stage. This stage includes image resizing, normalization, noise reduction, and contrast enhancement. Preprocessing improves image quality and reduces visual inconsistencies caused by lighting variations, camera angles, motion blur, and environmental conditions. As a result, the defect detection model receives cleaner and more standardized input data.

The proposed system uses YOLOv8 as the primary object detection algorithm because of its real-time detection capability, high accuracy, fast inference speed, and improved localization performance. YOLOv8 processes road images and identifies defect regions by generating bounding boxes, class labels, and confidence scores. The confidence score indicates the probability that a detected region belongs to a particular defect class. For example, if a pothole is detected with a confidence score of 94%, it indicates that the model is highly confident that the identified region represents a pothole.

Convolutional Neural Networks support the detection process by extracting meaningful visual features from road surface images. These features include edges, textures, shapes, shadow patterns, crack structures, and surface irregularities. Such feature extraction enables the system to distinguish between different types of road defects and normal road surface variations. OpenCV is used for computer vision operations such as image enhancement, filtering, contour analysis, and preprocessing support.

After detecting the defect type, the system estimates the severity level of the damage. The severity classification is categorized into four levels: Minor, Moderate, Major, and Critical. This classification is based on visual and structural indicators such as damaged area size, texture irregularities, edge sharpness, defect spread, shadow intensity, and surface deformation. Since normal two-dimensional images do not provide exact engineering-grade depth measurements, the system uses intelligent visual approximation techniques to estimate relative damage severity.

In addition to severity estimation, this module supports accident risk prediction. A Random Forest model is used to analyze defect-related features and predict risk levels such as Low Risk, Medium Risk, High Risk, and Critical Risk. The risk prediction process considers defect type, severity level, estimated damaged area, road condition, and environmental factors. This allows the system to prioritize defects that may pose greater safety risks to drivers and pedestrians.

V. Severity Analysis and Accident Risk Prediction

The severity analysis and accident risk prediction module is responsible for evaluating the seriousness of each detected

road defect and determining its potential impact on road safety. While defect detection identifies the type and location of pavement damage, severity and risk analysis provides additional decision-support information required for prioritizing maintenance activities.

The proposed system classifies road defect severity into four levels: Minor, Moderate, Major, and Critical. This classification is based on several visual and structural indicators extracted from the detected road damage region. These indicators include estimated damaged area, defect spread, edge sharpness, texture irregularities, shadow intensity, and surface deformation. For example, a small crack with limited spread may be classified as Minor, while a large pothole with sharp edges and high shadow intensity may be classified as Critical.

Since standard two-dimensional road images do not provide exact physical depth measurements, the system uses intelligent visual approximation techniques to estimate relative severity. Deep potholes and severe road surface failures often contain darker internal regions, stronger shadow patterns, sharper boundary transitions, and higher texture variation. These visual features are analyzed to support severity prediction without requiring expensive depth sensors such as LiDAR or stereo cameras.

Accident risk prediction is performed after the severity level has been estimated. The system predicts four accident risk levels: Low Risk, Medium Risk, High Risk, and Critical Risk. The risk prediction process considers the detected defect type, severity level, approximate damaged area, road surface condition, environmental conditions, and location-related information. For instance, a minor crack on a low-traffic road may be assigned a Low Risk level, whereas a major pothole located in a high-traffic lane may be classified as Critical Risk.

A Random Forest model is used for severity classification and accident risk prediction because it is effective for structured predictive analysis and can handle multiple input features. The model combines decision outputs from multiple trees to improve prediction stability and reduce overfitting. Input features may include defect category, confidence score, estimated area, texture-based indicators, shadow intensity, environmental condition, and road usage context. The final output is generated by aggregating the predictions from the decision trees within the model.

VI. Repair Area Estimation Using Semantic Segmentation

Accurate repair area estimation is a critical requirement for effective road maintenance planning and resource allocation. While traditional object detection algorithms can identify the presence and approximate location of road defects, they are generally limited to rectangular bounding boxes that do not accurately represent the irregular shapes of damaged regions. To overcome this limitation, the proposed Road Inspector system utilizes semantic segmentation techniques to perform pixel-level classification of road defects and estimate precise repair boundaries.

The primary objective of this module is to identify the exact damaged region requiring repair and convert the segmented defect area into approximate real-world measurements. This enables maintenance authorities to accurately determine the extent of road damage and estimate the resources required for repair operations. The module employs YOLOv8 Segmentation and U-Net architecture to achieve high-precision road damage segmentation.

YOLOv8 Segmentation is used to generate segmentation masks for detected road defects. Unlike conventional object detection models that produce only bounding boxes, YOLOv8 Segmentation identifies the actual shape of the damaged region by assigning a class label to each relevant pixel. This enables more accurate extraction of defect boundaries and provides a detailed representation of the damaged surface area.

To further improve segmentation accuracy, the system incorporates the U-Net architecture. U-Net is a deep learning-based semantic segmentation model that has demonstrated excellent performance in infrastructure monitoring and defect analysis applications. The encoder-decoder structure of U-Net enables efficient feature extraction while preserving spatial information required for accurate localization of damaged regions. This architecture is particularly effective for segmenting irregularly shaped defects such as potholes, cracks, and surface erosion.

The repair area estimation workflow begins with the segmented output generated by YOLOv8 Segmentation and U-Net. The damaged region is isolated from the surrounding road surface and converted into a binary damage mask. The system then counts the number of pixels belonging to the segmented defect region. Since pixel measurements alone do not provide meaningful physical dimensions, pixel-to-real-world calibration techniques are applied to estimate actual repair boundaries.

Pixel calibration is performed using reference scaling techniques that establish an approximate relationship between image pixels and real-world dimensions. For example, if calibration determines that 15 pixels correspond to 1 centimeter, the total segmented pixels can be converted into estimated width, length, and repair area measurements. This approach enables the system to estimate damaged surface dimensions without requiring specialized hardware such as LiDAR sensors or stereo vision cameras.

VII. Intelligent Material and Heat Estimation

Accurate estimation of repair materials and heating requirements is essential for efficient road maintenance planning, cost reduction, and resource optimization. Traditional material estimation methods are often performed manually and rely heavily on engineering experience, resulting in inconsistent calculations, material wastage, inaccurate budgeting, and inefficient repair operations. To address these challenges, the proposed Road Inspector system incorporates an Intelligent Material and Heat Estimation Module capable of automatically calculating repair material requirements and optimized heating conditions based on detected road damage characteristics.

The primary objective of this module is to estimate the quantity of asphalt, aggregate materials, tar requirements, repair volume, and heating temperature ranges required for effective road maintenance. The module receives repair area measurements generated by the semantic segmentation module

and combines them with severity information, environmental conditions, and road surface characteristics to produce intelligent maintenance recommendations.

The proposed system utilizes Random Forest algorithms and predictive estimation techniques to perform material requirement calculations. Random Forest was selected because of its strong predictive capabilities, ability to handle multiple input variables, and robustness when processing structured maintenance data. The model analyzes features such as defect type, estimated repair area, severity level, road condition, environmental conditions, and historical maintenance information to generate material requirement predictions.

The material estimation process begins by receiving the estimated repair area from the segmentation module. The system then evaluates the size and severity of the damaged region and calculates the approximate volume of material required for repair. Based on this information, the module estimates asphalt quantity, aggregate requirements, tar consumption, and overall repair material volume. These estimates assist maintenance authorities in procurement planning and budget allocation while minimizing unnecessary material wastage.

In addition to material estimation, the proposed system performs intelligent heat estimation to determine suitable heating temperature ranges for asphalt preparation and application. Heating temperature plays a significant role in repair quality, durability, and energy consumption. Several factors influence the heat estimation process, including environmental temperature, humidity levels, weather conditions, repair area size, estimated defect severity, and road surface characteristics.

VIII. Environmental and Traffic Analysis

Environmental conditions and traffic patterns play a significant role in determining the effectiveness, durability, and scheduling of road maintenance activities. Repairs performed under unfavorable weather conditions or during periods of high traffic congestion may result in reduced repair quality, increased operational costs, and disruptions to road users. To address these challenges, the proposed Road Inspector system incorporates an Environmental and Traffic Analysis Module that supports intelligent maintenance planning and decision-making.

The primary objective of this module is to analyze environmental factors and traffic conditions to identify optimal periods for road repair operations. By incorporating predictive analytics and environmental intelligence, the system assists maintenance authorities in selecting suitable repair schedules while minimizing disruptions and maximizing repair effectiveness.

The environmental analysis component evaluates several weather-related factors, including temperature, humidity, rainfall probability, and general weather suitability. These factors directly influence the performance of asphalt materials and repair processes. For example, excessive moisture or rainfall may weaken asphalt bonding, reduce repair durability, and increase the likelihood of maintenance failure. Similarly, extreme temperatures may affect material handling and curing conditions.

In addition to weather analysis, the system performs traffic analysis to evaluate road usage patterns and congestion levels. Traffic information is obtained through GPS integration, mapping services, and predictive traffic analysis techniques. The system identifies peak and off-peak traffic periods and recommends repair schedules that minimize traffic disruption and improve operational efficiency.

TABLE I
Road Defect Detection Performance

Defect Type	Detection Accuracy (%)
Potholes	82.2
Longitudinal Cracks	75.8
Transverse Cracks	74.6
Alligator Cracks	73.4
Edge Failures	88.7
Surface Erosion	87.9

TABLE II
Repair Area Estimation Performance

Evaluation Metric	Value
Segmentation Accuracy	82.6%
Intersection over Union	78.4%
Pixel Classification Accuracy	81.7%
Boundary Detection Accuracy	79.2%

IX. Experimental Results and Discussion

The experimental evaluation assessed road defect detection and repair-area segmentation while considering the broader prediction and decision-support workflow.

A. Experimental Setup

The system was implemented using Python, YOLOv8, OpenCV, PyTorch, U-Net, Random Forest, FastAPI, React, and MongoDB Atlas. Road images containing potholes, longitudinal cracks, transverse cracks, alligator cracks, edge failures, and surface erosion were processed under different environmental and lighting conditions.

B. Road Defect Detection Performance

YOLOv8 detected multiple road-damage categories with real-time processing capability. CNN-based features supported separation of road defects from normal pavement variations. Table I summarizes the reported detection accuracy for the six defect classes.

C. Repair Area Estimation Performance

YOLOv8 Segmentation and U-Net generated pixel-level masks for damaged regions. The reported segmentation accuracy, Intersection over Union, pixel classification accuracy, and boundary detection accuracy are shown in Table II.

D. Discussion

The results indicate that Road Inspector can combine defect identification, severity analysis, accident risk, repair area estimation, material calculation, environmental intelligence, and scheduling within one ecosystem. Limitations remain: detection accuracy can be affected by lighting, camera position, image

quality, and dataset diversity, while area values depend on calibrated image approximation rather than direct physical measurement. Future improvements may use depth estimation, stereo vision, LiDAR, or drone-based inspection to improve accuracy and reliability.

X. Conclusion and Future Work

Road Inspector demonstrates an integrated AI-powered approach to road inspection and repair estimation. The framework combines YOLOv8, CNN-based processing, U-Net semantic segmentation, Random Forest predictive models, IoT-enabled data collection, and environmental analysis to automate defect detection, severity assessment, accident risk prediction, repair area estimation, material estimation, and maintenance scheduling.

The system reduces dependence on manual observation and provides a unified basis for road-condition analysis and maintenance decision support. Future work may incorporate stereo vision, LiDAR, 3D reconstruction, and drone-based inspection to improve area and volume estimation. Additional developments may include predictive road degradation, AI-based maintenance prioritization, smart-city integration, larger datasets, real-time government monitoring, traffic-management integration, and route optimization. Overall, the framework provides a foundation for safer and more data-driven road infrastructure management.

References

[1] M. Rathee et al., "Automated road defect and anomaly detection for traffic safety," *PMC Systematic Review*, 2023.

[2] A. Alrajhi, K. Roy, L. Qingge, and J. Kribs, "Detection of road condition defects using multiple sensors and IoT technology: A review," *IEEE Open Journal of Intelligent Transportation Systems*, 2023.

[3] S. Swetha and S. Devnath, "AI-assisted road defect detection using a deep learning image segmentation model," *International Journal of Research Publication and Reviews*, 2023.

[4] G. Kim et al., "A road defect detection system using smartphones," *PMC Journal*, 2024.

[5] Z. Chen et al., "RoadAtlas: Intelligent platform for automated road defect detection and asset management," *arXiv*, 2021.

[6] R. Mukherjee et al., "AI driven road maintenance inspection," *arXiv*, 2021.

[7] J. J. Yebes et al., "Learning to automatically catch potholes in worldwide road scene images," *AUTOPILOT H2020, arXiv*.

[8] Y. Maeda et al., "Deep learning-based road damage detection and classification system," in *Proc. IEEE International Conference on Big Data*, 2018.

[9] Z. Lei et al., "Vision-based road crack detection using deep convolutional neural networks," in *Proc. IEEE International Conference on Image Processing*.

[10] I. Katsamenis et al., "A deep learning framework for segmentation of road damage," *ACM Digital Library*, 2024.

[11] J. Fan et al., "Multi-scale feature fusion: Learning better semantic segmentation for road pothole detection," *arXiv*, 2021.

[12] T.-Y. Tseng et al., "M2S-RoAD: Multi-modal semantic segmentation for road damage using camera and LiDAR data," *arXiv*, 2025.

[13] B. Kulambayev et al., "Real-time road damage detection system on deep learning based image analysis," *International Journal of Advanced Computer Science and Applications*, 2024.

[14] H. Arya et al., "Road damage semantic segmentation with deep learning," in *Proc. IEEE International Conference on Computing and Communication*.

[15] F. Bastani et al., "DeepRoadMapper: Extracting road topology from aerial images," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*.

[16] S. P. Kumar et al., "Semantic segmentation of road defects on pavement images using U-Net architecture," in *Proc. International Conference on Artificial Intelligence and Smart Systems*.

[17] Y. Fei et al., "Road crack segmentation using deep encoder-decoder network," *IEEE Access*.

[18] U. Zaimis, R. Jurmalietis, A. Jansone, and K. Kuduma, "Use of drone based IT system for road pothole detection and volume calculation," *International Scientific Journal Industry 4.0*, 2019.

[19] S. Bruno et al., "A small robot to repair asphalt road potholes," *Infrastructures*, 2024.

[20] G. S. Devi, D. V. R. Mohan, D. S. Reddy, et al., "Computer vision based volume estimation of potholes using ZED stereo camera," *International Journal of Recent Technology and Engineering*, 2019.

[21] J. Guan, X. Yang, L. Ding, X. Cheng, et al., "Automated pixel-level pavement distress detection based on stereo vision and deep learning," *Automation in Construction*, 2021.

[22] Y. E. P. R. Waliulu, P. Suprobo, and T. J. W. Adi, "Accuracy of hybrid models of detection, classification, and quantification for automatic road damage evaluation," *Civil Engineering and Architecture*, 2024.

[23] D. Wang, H. Zhu, Y. Xu, K. Liu, and R. Fan, "Robust video-based pothole detection and area estimation for intelligent vehicles with depth map and Kalman smoothing," *arXiv*, 2025.

[24] Q. Sun, L. Qiao, and Y. Shen, "Pavement potholes quantification: A study based on 3D point cloud analysis," *IEEE Access*, 2025.

[25] A. Wang, H. Lang, J. Chen, Z. Zou, et al., "Automated pavement pothole detection and multi-dimensional indicators extraction using deep learning and 3D reconstruction," *IEEE Transactions on Intelligent Transportation Systems*, 2026.

[26] A. Karukayil, C. Quail, and F. A. Cheein, "Deep learning enhanced feature extraction of potholes using vision and LiDAR data for road maintenance," *IEEE Access*, 2024.